

HOMS

Worksheet Memorandum

Motion in One Dimension

Physical Sciences | Grade 10 | Term 3

Marking guidance and worked answers. Total: 40 marks.

Learner name	
Date	

CAPS-aligned draft. Educator or subject-expert approval is required before classroom use.

Marking principle: Credit equivalent correct methods, units and directions. Do not award a final-answer mark when contradictory working is shown.

Question 1

- 1.1 An origin and a set of directions relative to which position and motion are described. [2]
- 1.2 Distance is total path length and is scalar. [2]
- Displacement is change in position and is vector.
- 1.3 The sign indicates direction relative to the chosen positive direction, not whether speed is decreasing. [2]
- 1.4 Velocity. [1]
- 1.5 Displacement. [1]

Question 2

- 2.1 Distance = $60 + 25 = 85$ m. [4]
- Displacement = $+60 - 25 = +35$ m, or 35 m east.
- 2.2 Average speed = $85/50 = 1.70$ m.s⁻¹. [4]
- Average velocity = $35/50 = 0.70$ m.s⁻¹ east.
- 2.3 $a = (16 - 4)/6 = +2.0$ m.s⁻². [3]
- 2.4 Take forward as positive. $v = 10 + (-2)(3) = 4$ m.s⁻¹ forward. [3]

Question 3

- 3.1 $v = \text{gradient} = (8 - 0)/(4 - 0) = +2$ m.s⁻¹. [3]
- 3.2 The object is at rest at position 8 m because the graph is horizontal. [2]
- 3.3 $v = (2 - 8)/(8 - 6) = -3$ m.s⁻¹. [3]
- 3.4 The object moves in the negative direction, back toward the origin. [2]

Question 4

- 4.1 $v = 3 + (2)(5) = 13$ m.s⁻¹. [3]
- 4.2 $\Delta x = (3)(5) + 0.5(2)(5^2) = 15 + 25 = 40$ m. [5]

TOTAL: 40